



ТЕХНИЧЕСКИ УНИВЕРСИТЕТ – ВАРНА

Факултет по изчислителна техника и автоматизация
Катедра „СИТ“

СЕМЕСТРИАЛНА ДОМАШНА РАБОТА

по дисциплината Базисни данни
на тема: Автобусни превози
Вариант 3

Изготвил: Теодора Стойчева

Проверил:

Специалност: СИТ

Група: 4-б

Факултетен номер: 22621693

2023

Задание

Да се проектира и реализира база от данни за автобусни превози, която да съхранява следната информация:

- Служител – име, позиция, телефон
- Клиент – име, телефон
- Билет – вид, дестинация, цена
- Резервации – дестинация, дата, час, цена, служител, клиент

Правила:

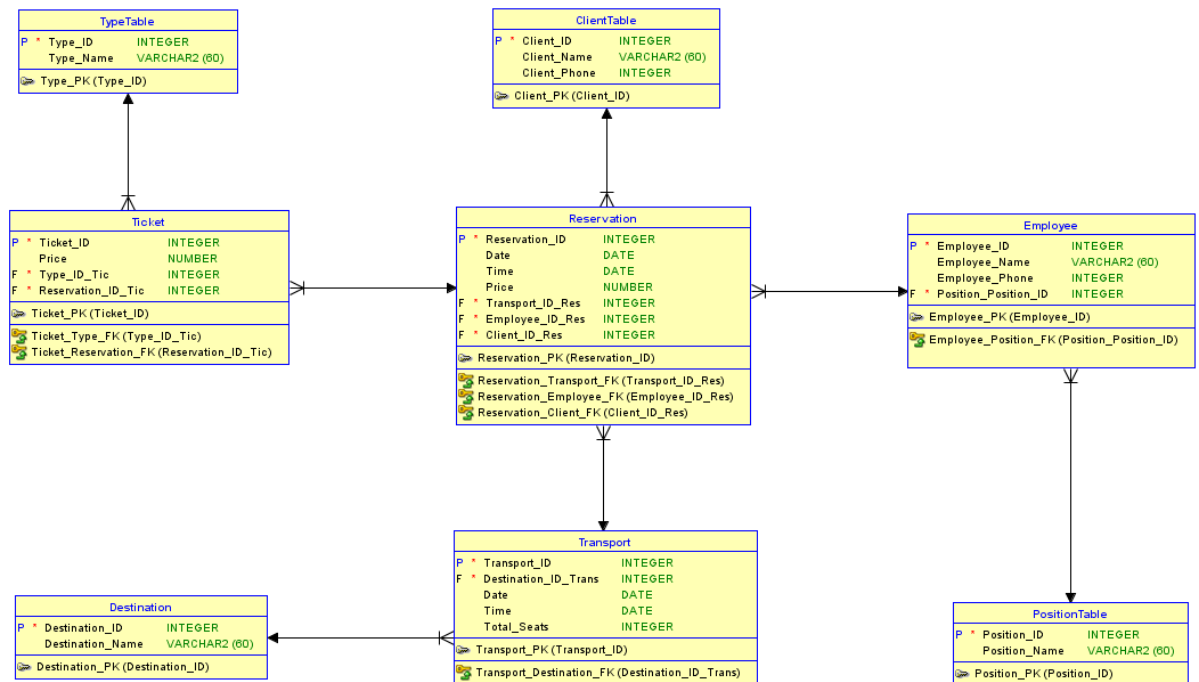
- Всеки клиент може да закупи или резервира повече от един билет
- По всяка дестинация има много превози

Базата от данни трябва да е нормализирана и да позволява:

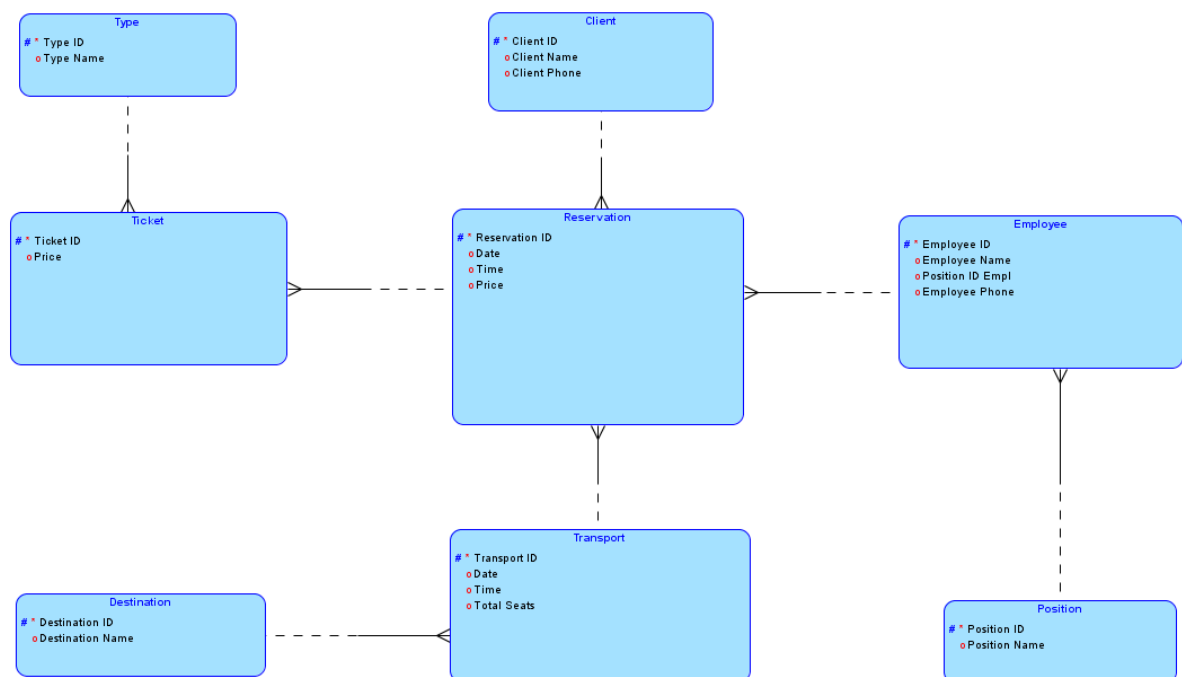
1. Въвеждане и корекции на данни
2. Търсене/резервации/закупуване на билети по: дестинация, дата, час.
3. Справки за продадени билети за клиент;
4. Справки за свободни места за дестинация/дата/час;
5. Справки за продажби за период, подредени по дати.

3.Модели

3.1 Релационен модел



3.2 Логически модел



4.DDL script (DROP,CREATE,ALTER TABLE)

CLIENTTABLE:

```
CREATE TABLE clienttable (  
    client_id    INTEGER NOT NULL,  
    client_name  VARCHAR2(60),  
    client_phone INTEGER  
);  
  
ALTER TABLE clienttable ADD CONSTRAINT client_pk  
PRIMARY KEY ( client_id );
```

DESTINATION:

```
CREATE TABLE destination (  
    destination_id    INTEGER NOT NULL,  
    destination_name  VARCHAR2(60)  
);  
  
ALTER TABLE destination ADD CONSTRAINT destination_pk  
PRIMARY KEY ( destination_id );
```

EMPLOYEE:

```
CREATE TABLE employee (  
    employee_id      INTEGER NOT NULL,  
    employee_name     VARCHAR2(60),  
    employee_phone    INTEGER,  
    position_position_id  INTEGER NOT NULL  
);  
  
ALTER TABLE employee ADD CONSTRAINT employee_pk  
PRIMARY KEY ( employee_id );
```

POSITIONTABLE:

```
CREATE TABLE positiontable (  
    position_id      INTEGER NOT NULL,  
    position_name     VARCHAR2(60)  
);  
  
ALTER TABLE positiontable ADD CONSTRAINT position_pk  
PRIMARY KEY ( position_id );
```

RESERVATION:

```
CREATE TABLE reservation (  
    reservation_id    INTEGER NOT NULL,  
    "Date"           DATE,  
    time             DATE,  
    price            NUMBER,  
    transport_id_res  INTEGER NOT NULL,  
    employee_id_res   INTEGER NOT NULL,  
    client_id_res     INTEGER NOT NULL  
);  
  
ALTER TABLE reservation ADD CONSTRAINT reservation_pk  
PRIMARY KEY ( reservation_id );
```

TICKET:

```
CREATE TABLE ticket (  
    ticket_id         INTEGER NOT NULL,  
    price            NUMBER,  
    type_id_tic       INTEGER NOT NULL,  
    reservation_id_tic INTEGER NOT NULL  
);
```

```
ALTER TABLE ticket ADD CONSTRAINT ticket_pk PRIMARY KEY  
( ticket_id );
```

TRANSPORT:

```
CREATE TABLE transport (  
    transport_id      INTEGER NOT NULL,  
    destination_id_trans INTEGER NOT NULL,  
    "Date"           DATE,  
    time             DATE,  
    total_seats       INTEGER  
);
```

```
ALTER TABLE transport ADD CONSTRAINT transport_pk  
PRIMARY KEY ( transport_id );
```

TYPETABLE:

```
CREATE TABLE typetable (  
    type_id  INTEGER NOT NULL,  
    type_name VARCHAR2(60)  
);
```

```
ALTER TABLE typetable ADD CONSTRAINT type_pk PRIMARY  
KEY ( type_id );
```

ALTER:

```
ALTER TABLE employee
```

```
ADD CONSTRAINT employee_position_fk FOREIGN KEY (  
position_position_id )
```

```
REFERENCES positiontable ( position_id );
```

```
ALTER TABLE reservation
```

```
ADD CONSTRAINT reservation_client_fk FOREIGN KEY (  
client_id_res )
```

```
REFERENCES clienttable ( client_id );
```

```
ALTER TABLE reservation
```

```
ADD CONSTRAINT reservation_employee_fk FOREIGN KEY (  
employee_id_res )
```

```
REFERENCES employee ( employee_id );
```

```
ALTER TABLE reservation
```

```
ADD CONSTRAINT reservation_transport_fk FOREIGN KEY (  
transport_id_res )
```

```
REFERENCES transport ( transport_id );
```



```
ALTER TABLE ticket
```

```
ADD CONSTRAINT ticket_reservation_fk FOREIGN KEY (
reservation_id_tic )
```

```
REFERENCES reservation ( reservation_id );
```

```
ALTER TABLE ticket
```

```
ADD CONSTRAINT ticket_type_fk FOREIGN KEY (
type_id_tic )
```

```
REFERENCES typetable ( type_id );
```

```
ALTER TABLE transport
```

```
ADD CONSTRAINT transport_destination_fk FOREIGN KEY (
destination_id_trans )
```

```
REFERENCES destination ( destination_id );
```

5. DML commands(INSERT,UPDATE,DELETE)

INSERT

```
INSERT INTO TYPETABLE (TYPE_ID, TYPE_NAME) VALUES (1, 'One Way');
INSERT INTO TYPETABLE (TYPE_ID, TYPE_NAME) VALUES (2, 'Round Trip');
```

```
Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(1,'Anna Ivanova',359001);
```

```
Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(2,'Ivan Ivanov',359002);
```

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(3,'Maria Peneva',359003);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(4,'Georgi Petkov',359004);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(5,'Georgi Georgiev',359005);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(6,'Stanislav Stanislavov',359006);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(7,'Aleksandra Ilieva',359007);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(8,'Martin Ganev',359008);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(9,'Ivanka Stoilova',359009);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(10,'Dimitar Petrov',359010);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(11,'Petar Dimitrov',359011);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(12,'Stoil Ivanov',359012);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(13,'Todor Yordanov',359013);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(14,'Boris Marinov',359014);

Insert into CLIENTTABLE (CLIENT_ID,CLIENT_NAME,CLIENT_PHONE) values
(15,'Nikola Velikov',359015);

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(1,'Sofia');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(2,'Varna');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(3,'Shumen');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(4,'Burgas');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(5,'Yambol');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(6,'Lovech');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(7,'Silistra');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(8,'Sliven');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(9,'Ruse');

Insert into DESTINATION (DESTINATION_ID,DESTINATION_NAME) values
(10,'Popovo');

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (1,'Teodor Todorov',359991,1);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (2,'Yordan Penev',359992,1);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (3,'Dimitar Nikolov',359993,1);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (4,'Stoyan Totev',359994,2);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (5,'Georgi Yordanov',359995,2);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (6,'Vasil Stoyanov',359996,2);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (7,'Krasimir Vasilev',359997,3);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (8,'Tanya Krasimirova',359998,4);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (9,'Velichka Teneva',359999,4);

Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (10,'Presiyan Velikov',359910,4);

```

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (11,'Maria Getseva',359911,5);

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (12,'Nikolina Marinova',359912,5);

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (13,'Elisaveta Dimcheva',359913,6);

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (14,'Kristiyan Dragomirov',359914,7);

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (15,'Ekaterina Boyanova',359915,7);

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (16,'Boyan Darinov',359916,8);

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (17,'Daria Radichkova',359917,9);

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (18,'Raya Dobromirova',359918,10);

Insert                                into                                EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (19,'Dobromir Hristov',359919,10);
Insert into EMPLOYEE
(EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID)
values (20,'Hristo Kozarev',359920,10);

```

Columns Data Model Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL				
	E...	EMPLOYEE_NAME	EMPLOYEE_PHONE	POSITION...
1	1	Teodor Todorov	359991	1
2	2	Yordan Penev	359992	1
3	3	Dimitar Nikolov	359993	1
4	4	Stoyan Totev	359994	2
5	5	Georgi Yordanov	359995	2
6	6	Vasil Stoyanov	359996	2
7	7	Krasimir Vasilev	359997	3
8	8	Tanya Krasimirova	359998	4
9	9	Velichka Teneva	359999	4
10	10	Presiyan Velikov	359910	4
11	11	Maria Getseva	359911	5
12	12	Nikolina Marinova	359912	5
13	13	Elisaveta Dimcheva	359913	6
14	14	Kristiyan Dragomirov	359914	7
15	15	Ekaterina Boyanova	359915	7
16	16	Boyan Darinov	359916	8
17	17	Daria Radichkova	359917	9
18	18	Raya Dobromirova	359918	10
19	19	Dobromir Hristov	359919	10
20	20	Hristo Kozarev	359920	10

Worksheet | Query Builder

1 | `Insert into EMPLOYEE (EMPLOYEE_ID,EMPLOYEE_NAME,EMPLOYEE_PHONE,POSITION_POSITION_ID) values (21,'Proba Proba',123,2);`

Script Output x

Task completed in 0,051 seconds

1 row inserted.

Columns Data Model Constraints Grants Statistics Triggers Flashback Dependencies Details Partitions Indexes SQL				
	E...	EMPLOYEE_NAME	EMPLOYEE_PHONE	POSITION...
1	1	Teodor Todorov	359991	1
2	2	Yordan Penev	359992	1
3	3	Dimitar Nikolov	359993	1
4	4	Stoyan Totev	359994	2
5	5	Georgi Yordanov	359995	2
6	6	Vasil Stoyanov	359996	2
7	7	Krasimir Vasilev	359997	3
8	8	Tanya Krasimirova	359998	4
9	9	Velichka Teneva	359999	4
10	10	Presiyan Velikov	359910	4
11	11	Maria Getseva	359911	5
12	12	Nikolina Marinova	359912	5
13	13	Elisaveta Dimcheva	359913	6
14	14	Kristiyan Dragomirov	359914	7
15	15	Ekaterina Boyanova	359915	7
16	16	Boyan Darinov	359916	8
17	17	Daria Radichkova	359917	9
18	18	Raya Dobromirova	359918	10
19	19	Dobromir Hristov	359919	10
20	20	Hristo Kozarev	359920	10
21	21	Proba Proba	123	2

Messages - Log

INSERT INTO "SYSTEM"."EMPLOYEE" (EMPLOYEE_ID, EMPLOYEE_NAME, EMPLOYEE_PHONE, POSITION_POSITION_ID) VALUES ('21', 'Proba Proba', '123', '2')
 Commit Successful

DESTINATION_ID	DESTINATION_NAME
1	Sofia
2	Varna
3	Shumen
4	Burgas
5	Yambol
6	Lovech
7	Silistra
8	Sliven
9	Ruse
10	Popovo

CLIENT_ID	CLIENT_NAME	CLIENT_PHONE
1	Anna Ivanova	359001
2	Ivan Ivanov	359002
3	Maria Peneva	359003
4	Georgi Petkov	359004
5	Georgi Georgiev	359005
6	Stanislav Stanislavov	359006
7	Aleksandra Ilieva	359007
8	Martin Ganev	359008
9	Ivanka Stoilova	359009
10	Dimitar Petrov	359010
11	Petar Dimitrov	359011
12	Stoil Ivanov	359012
13	Todor Yordanov	359013
14	Boris Marinov	359014
15	Nikola Velikov	359015

UPDATE

```
UPDATE CLIENTTABLE
SET CLIENT_NAME='Anna Ivanova' CLIENT_PHONE='359001'
WHERE CLIENT_ID=1;
```

```
UPDATE CLIENTTABLE
SET CLIENT_NAME='&CLIENT_NAME',CLIENT_PHONE='&CLIENT_PHONE'
WHERE CLIENT_ID=&CLIENT_ID;
```

```
UPDATE EMPLOYEE
SET EMPLOYEE_NAME'&EMPLOYEE_NAME'
WHERE EMPLOYEE_ID=&EMPLOYEE_ID;
```

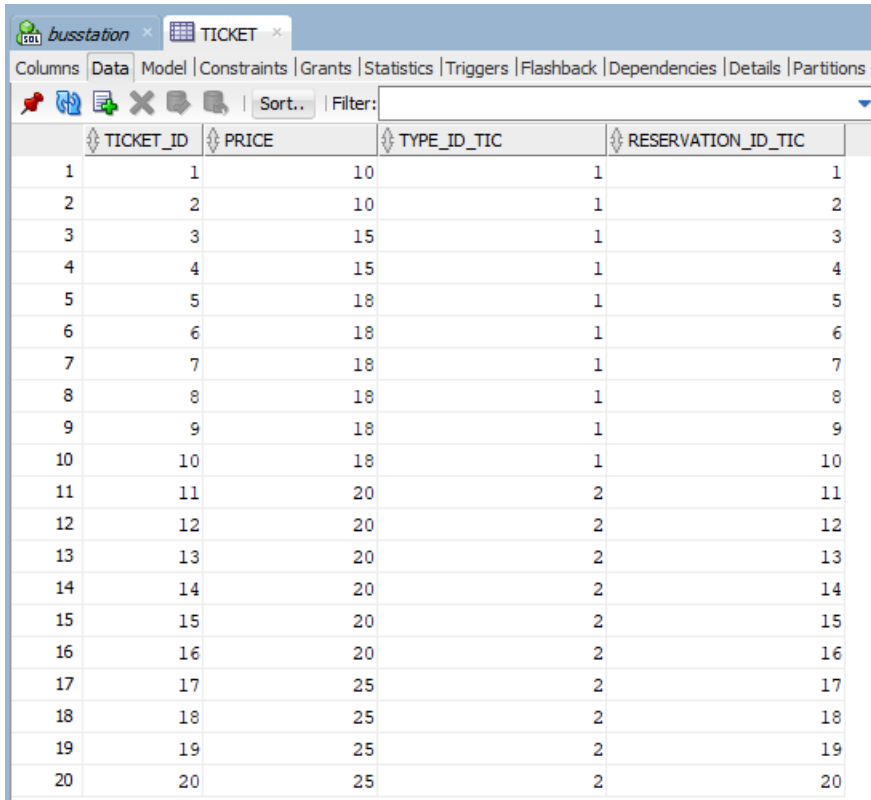
```
UPDATE EMPLOYEE
SET EMPLOYEE_PHONE='&EMPLOYEE_PHONE'
```

WHERE EMPLOYEE_ID=1

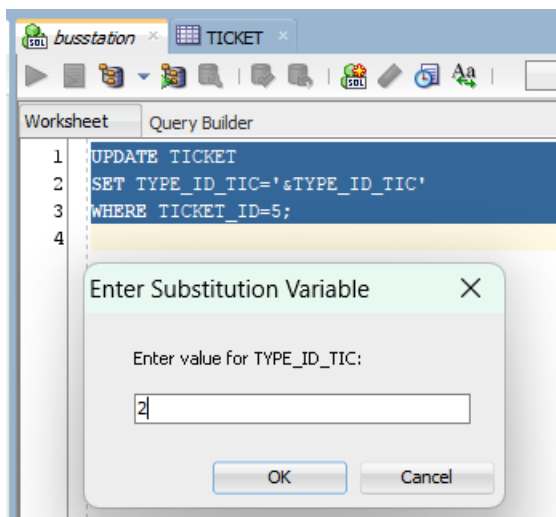
UPDATE TICKET

SET TYPE_ID_TIC='&TYPE_ID_TIC'

WHERE TICKET_ID=5;



TICKET_ID	PRICE	TYPE_ID_TIC	RESERVATION_ID_TIC
1	1	10	1
2	2	10	1
3	3	15	1
4	4	15	1
5	5	18	1
6	6	18	1
7	7	18	1
8	8	18	1
9	9	18	1
10	10	18	1
11	11	20	2
12	12	20	2
13	13	20	2
14	14	20	2
15	15	20	2
16	16	20	2
17	17	25	2
18	18	25	2
19	19	25	2
20	20	25	2



Worksheet	Query Builder
1	UPDATE TICKET
2	SET TYPE_ID_TIC='&TYPE_ID_TIC'
3	WHERE TICKET_ID=5;
4	

Script Output x

Task completed in 17,764 s

old:UPDATE TICKET
SET TYPE_ID_TIC='&TYPE_ID_TIC'
WHERE TICKET_ID=5
new:UPDATE TICKET
SET TYPE_ID_TIC='2'
WHERE TICKET_ID=5

1 row updated.

TICKET_ID	PRICE	TYPE_ID_TIC	RESERVATION_ID_TIC
1	1	10	1
2	2	10	1
3	3	15	1
4	4	15	1
5	5	18	2
6	6	18	1
7	7	18	1
8	8	18	1
9	9	18	1
10	10	18	1
11	11	20	2
12	12	20	2
13	13	20	2
14	14	20	2
15	15	20	2
16	16	20	2
17	17	25	2
18	18	25	2
19	19	25	2
20	20	25	2

DELETE

DELETE FROM DESTINATION
WHERE DESTINATION_ID=&DESTINATION_ID;

DELETE FROM TICKET
WHERE TICKET_ID=7;

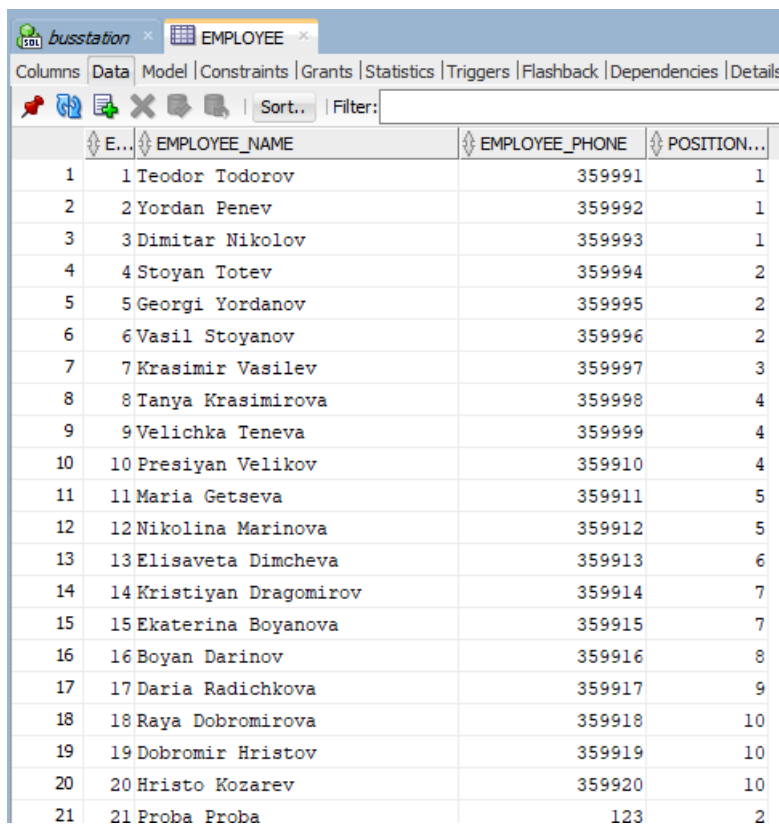
DELETE FROM CLIENT
WHERE CLIENT_ID=9;


```
DELETE FROM RESERVATION
WHERE TRANSPORT_ID_RES=&TRANSPORT_ID_RES;
```

```
DELETE FROM POSITIONTABLE
WHERE POSITION_ID=3;
```

```
DELETE FROM TRANSPORT
WHERE DESTINATION_ID_TRANS=3;
```

```
DELETE FROM EMPLOYEE
WHERE EMPLOYEE_ID=&EMPLOYEE_ID;
```



The screenshot shows a database management tool interface with a tab labeled 'EMPLOYEE'. The table data is displayed in a grid with columns: EMPLOYEE_ID, EMPLOYEE_NAME, EMPLOYEE_PHONE, and POSITION_ID. The data is as follows:

EMPLOYEE_ID	EMPLOYEE_NAME	EMPLOYEE_PHONE	POSITION_ID
1	Teodor Todorov	359991	1
2	Yordan Penev	359992	1
3	Dimitar Nikolov	359993	1
4	Stoyan Totev	359994	2
5	Georgi Yordanov	359995	2
6	Vasil Stoyanov	359996	2
7	Krasimir Vasilev	359997	3
8	Tanya Krasimirova	359998	4
9	Velichka Teneva	359999	4
10	Presiyan Velikov	359910	4
11	Maria Getseva	359911	5
12	Nikolina Marinova	359912	5
13	Elisaveta Dimcheva	359913	6
14	Kristiyan Dragomirov	359914	7
15	Ekaterina Boyanova	359915	7
16	Boyan Darinov	359916	8
17	Daria Radichkova	359917	9
18	Raya Dobromirova	359918	10
19	Dobromir Hristov	359919	10
20	Hristo Kozarev	359920	10
21	Proba Proba	123	2

The screenshot displays a database management interface with the following components:

- Query Builder:** Contains a SQL query:


```
1 DELETE FROM EMPLOYEE
2 WHERE EMPLOYEE_ID=&EMPLOYEE_ID;
3
```
- Script Output:** Shows the execution result:


```
old:DELETE FROM EMPLOYEE
WHERE EMPLOYEE_ID=&EMPLOYEE_ID
new:DELETE FROM EMPLOYEE
WHERE EMPLOYEE_ID=21
1 row deleted.
```
- Enter Substitution Variable:** A dialog box prompts for a value for EMPLOYEE_ID, with '21' entered.
- Table View:** A table with columns EMPLOYEE_ID, EMPLOYEE_NAME, EMPLOYEE_PHONE, and POSITION_ID. It contains 20 rows of employee data.

EMPLOYEE_ID	EMPLOYEE_NAME	EMPLOYEE_PHONE	POSITION_ID
1	Teodor Todorov	359991	1
2	Yordan Penev	359992	1
3	Dimitar Nikolov	359993	1
4	Stoyan Totev	359994	2
5	Georgi Yordanov	359995	2
6	Vasil Stoyanov	359996	2
7	Krasimir Vasilev	359997	3
8	Tanya Krasimirova	359998	4
9	Velichka Teneva	359999	4
10	Presiyan Velikov	359910	4
11	Maria Getseva	359911	5
12	Nikolina Marinova	359912	5
13	Elisaveta Dimcheva	359913	6
14	Kristiyan Dragomirov	359914	7
15	Ekaterina Boyanova	359915	7
16	Boyan Darinov	359916	8
17	Daria Radichkova	359917	9
18	Raya Dobromirova	359918	10
19	Dobromir Hristov	359919	10
20	Hristo Kozarev	359920	10

SELECT Търсене/резервиране/закупуване на билети по дестинация, дата, час

1.1 Дестинация:

SELECT

SELECT

```
r."Date" as reservation_date,  
r.time as reservation_time,  
t.type_name as ticket_type,  
c.client_name as client_name,  
tr.total_seats as total_seats
```

FROM ticket tic

JOIN reservation r ON tic.reservation_id_tic = r.reservation_id

JOIN typetable t ON tic.type_id_tic = t.type_id

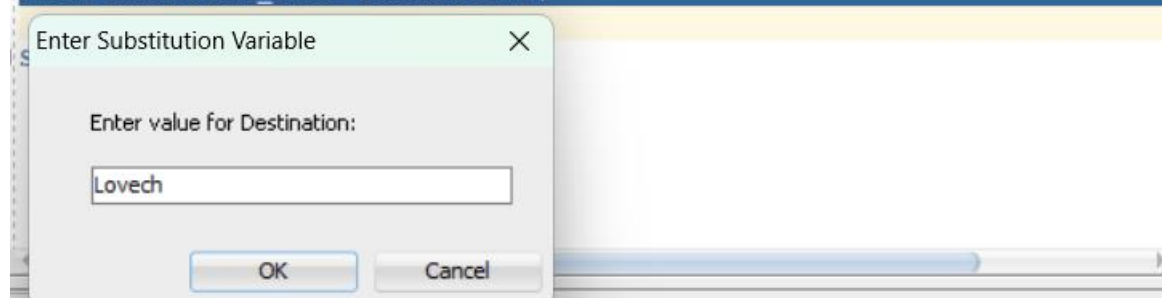
JOIN clienttable c ON r.client_id_res = c.client_id

JOIN transport tr ON r.transport_id_res = tr.transport_id

JOIN destination d ON tr.destination_id_trans = d.destination_id

WHERE destination_name='&Destination';

```
SELECT  
    r."Date" as reservation_date,  
    r.time as reservation_time,  
    t.type_name as ticket_type,  
    c.client_name as client_name,  
    tr."Date" as transport_date,  
    tr.time as transport_time,  
    tr.total_seats as total_seats  
FROM ticket tic  
JOIN reservation r ON tic.reservation_id_tic = r.reservation_id  
JOIN typetable t ON tic.type_id_tic = t.type_id  
JOIN clienttable c ON r.client_id_res = c.client_id  
JOIN transport tr ON r.transport_id_res = tr.transport_id  
JOIN destination d ON tr.destination_id_trans = d.destination_id  
WHERE destination_name='&Destination';
```



L446	SELECT
L447	r."Date" as reservation_date,
L448	r.time as reservation_time,
L449	t.type_name as ticket_type,
L450	c.client_name as client_name,
L451	tr.total_seats as total_seats
L452	FROM ticket tic
L453	JOIN reservation r ON tic.reservation_id_tic = r.reservation_id
L454	JOIN typetable t ON tic.type_id_tic = t.type_id
L455	JOIN clienttable c ON r.client_id_res = c.client_id
L456	JOIN transport tr ON r.transport_id_res = tr.transport_id
L457	JOIN destination d ON tr.destination_id_trans = d.destination_id
L458	WHERE destination_name='&Destination';
L459	

Query Result x					
All Rows Fetched: 2 in 0,004 seconds					
	RESERVATION_DATE	RESERVATION_TIME	TICKET_TYPE	CLIENT_NAME	TOTAL_SEATS
1	2024-11-04	2024-11-04 12:30:00.000000000	Round Trip	Petar Dimitrov	50
2	2024-11-04	2024-11-04 12:30:00.000000000	Round Trip	Stoil Ivanov	50

1.2 Data:

SELECT

 t.type_name as ticket_type,
 c.client_name as client_name,
 d.destination_name as destination,
 tr."Date" as transport_date,
 tr.time as transport_time,
 tr.total_seats as total_seats

FROM ticket tic

JOIN reservation r ON tic.reservation_id_tic = r.reservation_id

JOIN typetable t ON tic.type_id_tic = t.type_id

JOIN clienttable c ON r.client_id_res = c.client_id

JOIN transport tr ON r.transport_id_res = tr.transport_id

JOIN destination d ON tr.destination_id_trans = d.destination_id

WHERE r."Date" = TO_DATE('&Date', 'DD-MM-YYYY');

busstation CLIENTTABLE

Worksheet Query Builder

```

1459
1460 SELECT
1461     t.type_name as ticket_type,
1462     c.client_name as client_name,
1463     d.destination_name as destination,
1464     tr."Date" as transport_date,
1465     tr.time as transport_time,
1466     tr.total_seats as total_seats
1467 FROM ticket tic
1468 JOIN reservation r ON tic.reservation_id_tic = r.reservation_id
1469 JOIN typetable t ON tic.type_id_tic = t.type_id
1470 JOIN clienttable c ON r.client_id_res = c.client_id
1471 JOIN transport tr ON r.transport_id_res = tr.transport_id
1472 JOIN destination d ON tr.destination_id_trans = d.destination_id
1473 WHERE r."Date" = TO_DATE('&Date', 'DD-MM-YYYY');
1474
1475

```

Enter Substitution Variable

Enter value for Date:

05-11-2024

OK Cancel

	TICKET_TYPE	CLIENT_NAME	TOTAL_SEATS
0.000000000	Round Trip	Petar Dimitrov	50
0.000000000	Round Trip	Stoil Ivanov	50

Query Result x

All Rows Fetched: 6 in 0,004 seconds

	TICKET_TYPE	CLIENT_NAME	DESTINATION	TRANSPORT_DATE	TRANSPORT_TIME	TOTAL_SEATS
1	Round Trip	Todor Yordanov	Silistra	2024-11-05	2024-11-05 10:00:00.000000000	50
2	Round Trip	Boris Marinov	Silistra	2024-11-05	2024-11-05 10:00:00.000000000	50
3	Round Trip	Nikola Velikov	Sliven	2024-11-05	2024-11-05 11:00:00.000000000	50
4	Round Trip	Anna Ivanova	Sliven	2024-11-05	2024-11-05 11:00:00.000000000	50
5	Round Trip	Ivan Ivanov	Ruse	2024-11-05	2024-11-05 12:00:00.000000000	60
6	Round Trip	Maria Peneva	Ruse	2024-11-05	2024-11-05 12:00:00.000000000	60

1.3 Чак:

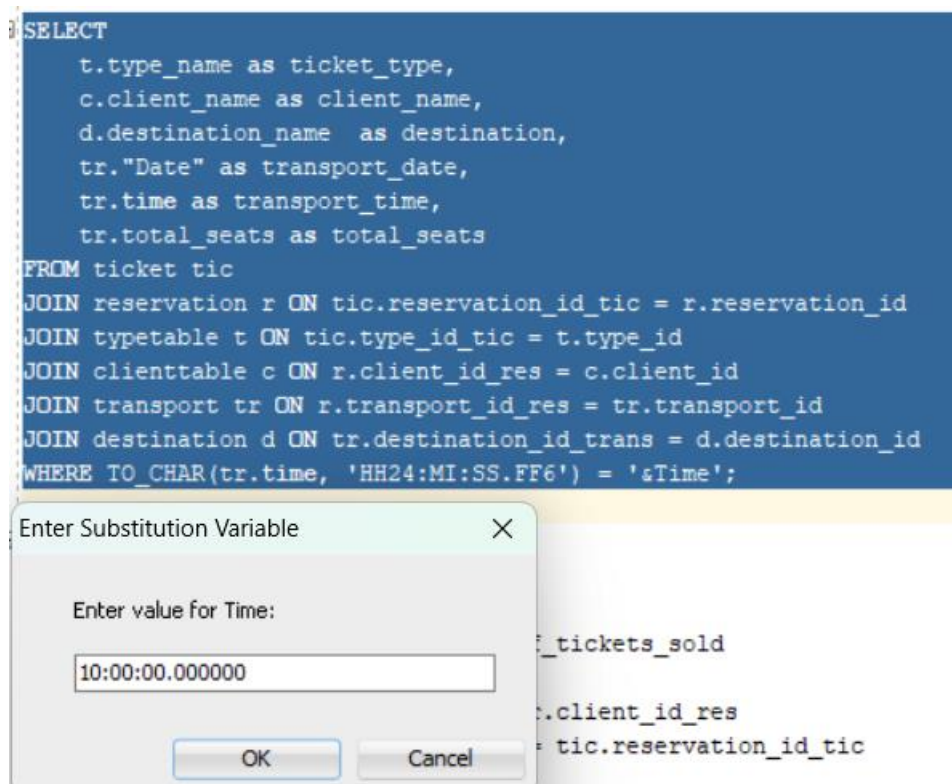
SELECT

t.type_name as ticket_type,
c.client_name as client_name,
d.destination_name as destination,

```

tr."Date" as transport_date,
tr.time as transport_time,
tr.total_seats as total_seats
FROM ticket tic
JOIN reservation r ON tic.reservation_id_tic = r.reservation_id
JOIN typetable t ON tic.type_id_tic = t.type_id
JOIN clienttable c ON r.client_id_res = c.client_id
JOIN transport tr ON r.transport_id_res = tr.transport_id
JOIN destination d ON tr.destination_id_trans = d.destination_id
WHERE TO_CHAR(tr.time, 'HH24:MI:SS.FF6') = '&Time';

```



Query Result x						
All Rows Fetched: 6 in 0 seconds						
	TICKET_TYPE	CLIENT_NAME	DESTINATION	TRANSPORT_DATE	TRANSPORT_TIME	TOTAL_SEATS
1	One Way	Maria Peneva	Varna	2024-11-02	2024-11-02 10:00:00.000000000	40
2	One Way	Georgi Petkov	Varna	2024-11-02	2024-11-02 10:00:00.000000000	40
3	Round Trip	Todor Yordanov	Silistra	2024-11-05	2024-11-05 10:00:00.000000000	50
4	Round Trip	Boris Marinov	Silistra	2024-11-05	2024-11-05 10:00:00.000000000	50
5	Round Trip	Georgi Petkov	Popovo	2024-11-06	2024-11-06 10:00:00.000000000	60
6	Round Trip	Georgi Georgiev	Popovo	2024-11-06	2024-11-06 10:00:00.000000000	60

Справки за:

Продадени билети на клиент:

```

SELECT
    c.client_name AS client_name,
    c.client_phone AS client_phone,
    COUNT(tic.ticket_id) AS number_of_tickets_sold
FROM clienttable c
JOIN reservation r ON c.client_id = r.client_id_res
JOIN ticket tic ON r.reservation_id = tic.reservation_id_tic
GROUP BY c.client_name, c.client_phone;

```


busstation x CLIENTTABLE x

Worksheet Query Builder

```

1490 SELECT
1491     c.client_name AS client_name,
1492     c.client_phone AS client_phone,
1493     COUNT(tic.ticket_id) AS number_of_tickets_sold
1494 FROM clienttable c
1495 JOIN reservation r ON c.client_id = r.client_id_res
1496 JOIN ticket tic ON r.reservation_id = tic.reservation_id_tic
1497 GROUP BY
1498     c.client_name,
1499     c.client_phone;

```

Query Result x

SQL | All Rows Fetched: 15 in 0 seconds

	CLIENT_NAME	CLIENT_PHONE	NUMBER_OF_TICKETS_SOLD
1	Georgi Petkov	359004	2
2	Todor Yordanov	359013	1
3	Georgi Georgiev	359005	2
4	Maria Peneva	359003	2
5	Ivanka Stoilova	359009	1
6	Martin Ganev	359008	1
7	Dimitar Petrov	359010	1
8	Boris Marinov	359014	1
9	Stanislav Stanislavov	359006	1
10	Nikola Velikov	359015	1
11	Ivan Ivanov	359002	2
12	Aleksandra Ilieva	359007	1
13	Stoil Ivanov	359012	1
14	Anna Ivanova	359001	2
15	Petar Dimitrov	359011	1

Свободни места за дестинация, дата, час

SELECT

d.destination_name AS destination_name,

tr."Date" AS transport_date,

tr.time AS transport_time,

tr.total_seats - COUNT(tic.ticket_id) AS number_of_free_seats

FROM transport tr

JOIN destination d ON tr.destination_id_trans = d.destination_id

LEFT JOIN reservation r ON r.transport_id_res = tr.transport_id

LEFT JOIN ticket tic ON r.reservation_id = tic.reservation_id_tic

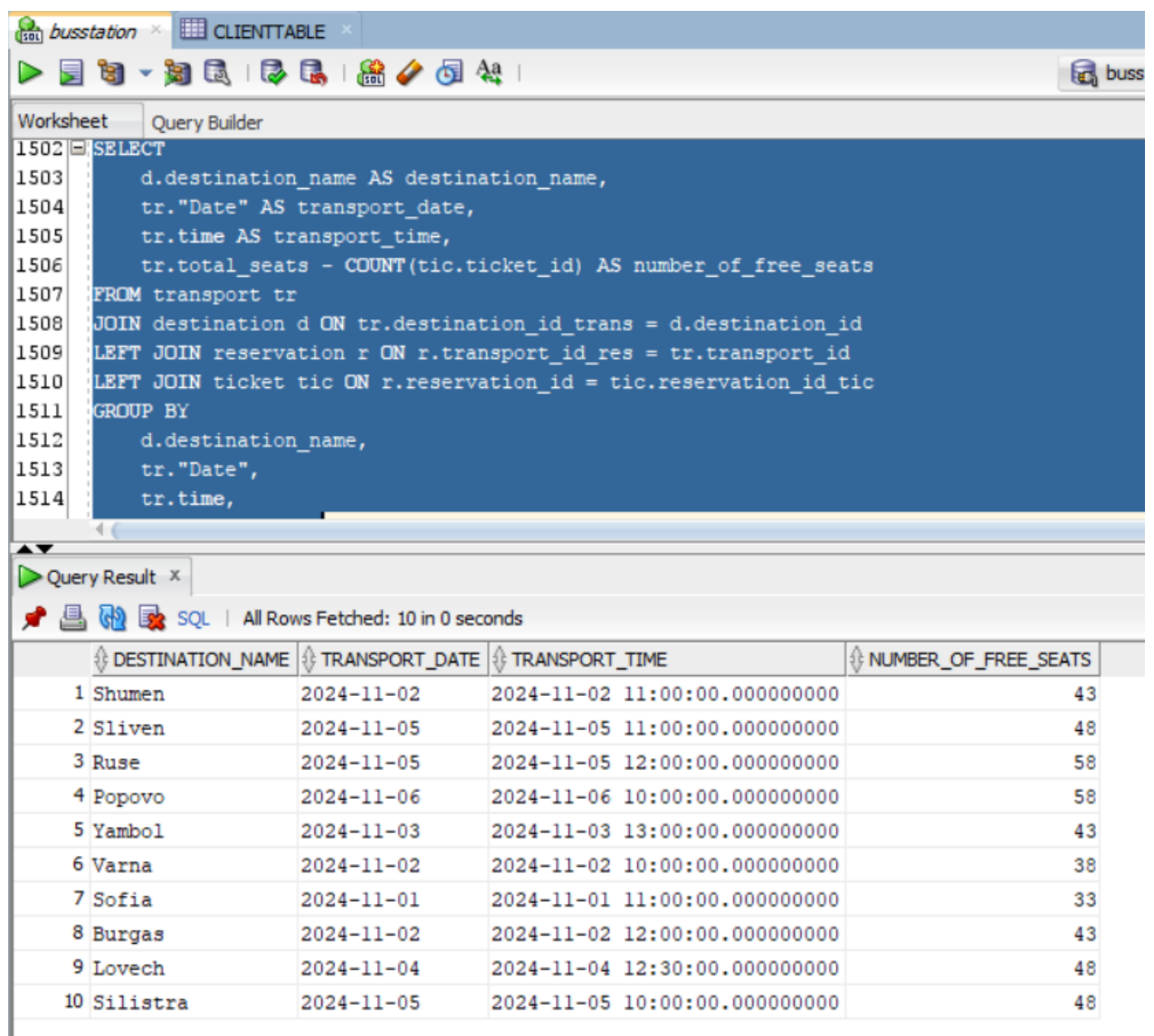
GROUP BY

d.destination_name,

tr."Date",

tr.time,

tr.total_seats;



The screenshot displays a database query builder interface for a project named 'busstation'. The query builder window shows the following SQL query:

```
SELECT
    d.destination_name AS destination_name,
    tr."Date" AS transport_date,
    tr.time AS transport_time,
    tr.total_seats - COUNT(tic.ticket_id) AS number_of_free_seats
FROM transport tr
JOIN destination d ON tr.destination_id_trans = d.destination_id
LEFT JOIN reservation r ON r.transport_id_res = tr.transport_id
LEFT JOIN ticket tic ON r.reservation_id = tic.reservation_id_tic
GROUP BY
    d.destination_name,
    tr."Date",
    tr.time,
```

The query results window shows the following table:

	DESTINATION_NAME	TRANSPORT_DATE	TRANSPORT_TIME	NUMBER_OF_FREE_SEATS
1	Shumen	2024-11-02	2024-11-02 11:00:00.000000000	43
2	Sliven	2024-11-05	2024-11-05 11:00:00.000000000	48
3	Ruse	2024-11-05	2024-11-05 12:00:00.000000000	58
4	Popovo	2024-11-06	2024-11-06 10:00:00.000000000	58
5	Yambol	2024-11-03	2024-11-03 13:00:00.000000000	43
6	Varna	2024-11-02	2024-11-02 10:00:00.000000000	38
7	Sofia	2024-11-01	2024-11-01 11:00:00.000000000	33
8	Burgas	2024-11-02	2024-11-02 12:00:00.000000000	43
9	Lovech	2024-11-04	2024-11-04 12:30:00.000000000	48
10	Silistra	2024-11-05	2024-11-05 10:00:00.000000000	48

Продажби за период, подредени по дати

SELECT

tr."Date" AS transport_date,
tr.time AS transport_time,
d.destination_name AS destination,
c.client_name AS client_name,
t.type_name AS ticket_type

FROM ticket tic

JOIN reservation r ON tic.reservation_id_tic = r.reservation_id

JOIN transport tr ON r.transport_id_res = tr.transport_id

JOIN destination d ON tr.destination_id_trans = d.destination_id

JOIN clienttable c ON r.client_id_res = c.client_id

JOIN typetable t ON tic.type_id_tic = t.type_id

WHERE tr."Date" BETWEEN TO_DATE('&StartDate', 'DD-MM-YYYY') AND
TO_DATE('&EndDate', 'DD-MM-YYYY')

ORDER BY tr."Date" ASC;

busstation CLIENTTABLE

Worksheet Query Builder

```

1517 SELECT
1518     tr."Date" AS transport_date,
1519     tr.time AS transport_time,
1520     d.destination_name AS destination,
1521     c.client_name AS client_name,
1522     t.type_name AS ticket_type
1523 FROM ticket tic
1524 JOIN reservation r ON tic.reservation_id_tic = r.reservation_id
1525 JOIN transport tr ON r.transport_id_res = tr.transport_id
1526 JOIN destination d ON tr.destination_id_trans = d.destination_id
1527 JOIN clienttable c ON r.client_id_res = c.client_id
1528 JOIN typetable t ON tic.type_id_tic = t.type_id
1529 WHERE tr."Date" BETWEEN TO DATE('&StartDate', 'DD-MM-YYYY') AND TO DATE('&EndDate', 'DD-MM-YYYY')
1530 ORDER BY tr."Date" ASC;

```

Enter Substitution Variable

Enter value for StartDate:

02-11-2024

OK Cancel

	DESTINATION	CLIENT_NAME	TICKET_TYPE	
0.000000000	Yambol	Ivanka Stoilova	One Way	
0.000000000	Yambol	Dimitar Petrov	One Way	
0.000000000	Lovech	Petar Dimitrov	Round Trip	
4 2024-11-04	2024-11-04 12:30:00.000000000	Lovech	Stoil Ivanov	Round Trip

busstation CLIENTTABLE

Worksheet Query Builder

```

1517 SELECT
1518     tr."Date" AS transport_date,
1519     tr.time AS transport_time,
1520     d.destination_name AS destination,
1521     c.client_name AS client_name,
1522     t.type_name AS ticket_type
1523 FROM ticket tic
1524 JOIN reservation r ON tic.reservation_id_tic = r.reservation_id
1525 JOIN transport tr ON r.transport_id_res = tr.transport_id
1526 JOIN destination d ON tr.destination_id_trans = d.destination_id
1527 JOIN clienttable c ON r.client_id_res = c.client_id
1528 JOIN typetable t ON tic.type_id_tic = t.type_id
1529 WHERE tr."Date" BETWEEN TO DATE('&StartDate', 'DD-MM-YYYY') AND TO DATE('&EndDate', 'DD-MM-YYYY')
1530 ORDER BY tr."Date" ASC;

```

Enter Substitution Variable

Enter value for EndDate:

05-11-2024

OK Cancel

TRANSPORT_TIME	NUMBER_OF_FREE_SEATS
24-11-02 11:00:00.000000000	43
24-11-05 11:00:00.000000000	48
24-11-05 12:00:00.000000000	58

Query Result x					
SQL All Rows Fetched: 16 in 0 seconds					
	TRANSPORT_DATE	TRANSPORT_TIME	DESTINATION	CLIENT_NAME	TICKET_TYPE
1	2024-11-02	2024-11-02 10:00:00.000000000	Varna	Maria Peneva	One Way
2	2024-11-02	2024-11-02 10:00:00.000000000	Varna	Georgi Petkov	One Way
3	2024-11-02	2024-11-02 11:00:00.000000000	Shumen	Georgi Georgiev	One Way
4	2024-11-02	2024-11-02 11:00:00.000000000	Shumen	Stanislav Stanislavov	One Way
5	2024-11-02	2024-11-02 12:00:00.000000000	Burgas	Aleksandra Ilieva	One Way
6	2024-11-02	2024-11-02 12:00:00.000000000	Burgas	Martin Ganev	One Way
7	2024-11-03	2024-11-03 13:00:00.000000000	Yambol	Ivanka Stoilova	One Way
8	2024-11-03	2024-11-03 13:00:00.000000000	Yambol	Dimitar Petrov	One Way
9	2024-11-04	2024-11-04 12:30:00.000000000	Lovech	Petar Dimitrov	Round Trip
10	2024-11-04	2024-11-04 12:30:00.000000000	Lovech	Stoil Ivanov	Round Trip
11	2024-11-05	2024-11-05 10:00:00.000000000	Silistra	Todor Yordanov	Round Trip
12	2024-11-05	2024-11-05 10:00:00.000000000	Silistra	Boris Marinov	Round Trip
13	2024-11-05	2024-11-05 11:00:00.000000000	Sliven	Nikola Velikov	Round Trip
14	2024-11-05	2024-11-05 11:00:00.000000000	Sliven	Anna Ivanova	Round Trip
15	2024-11-05	2024-11-05 12:00:00.000000000	Ruse	Ivan Ivanov	Round Trip
16	2024-11-05	2024-11-05 12:00:00.000000000	Ruse	Maria Peneva	Round Trip